

# VICTOR RODRIGUEZ

HPC research engineer

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[epsylene.dev](https://github.com/Epsylene)

## SUMMARY

HPC research engineer specialized in GPU computing and performance optimization. Master's degrees in numerical physics and graphics programming. Experienced in C++, Python, Kokkos, CUDA, and MPI, currently porting a climate model I/O library to GPU at CEA.

## SKILLS

**Languages:** C++, Rust, Python, Fortran, x86-64 asm

**Technologies:** Linux, Git, CMake, OpenMP, MPI, CUDA

## EXPERIENCE

- 3/2026 – ongoing **Research engineer** CEA/LSCE  
Working for the **CALCULS** team, focusing on rewriting the XIOS scientific I/O library with Kokkos to enable GPU support.
- 3/2025 – 8/2025 **M2 Internship** BUF Compagnie  
Development and integration of a large-scale PBD grain solver
- 2/2024 – 8/2024 **M2 Internship** CEA/LCAN  
Benchmarking and development of a Kokkos-based HPC GPU diffusion solver

## EDUCATION

- 9/2024 – 6/2025 **Master ID3D (Image, Développement et Technologies 3D)** University Claude Bernard Lyon 1 (France)  
Second year of Master's degree in informatics, specializing in graphics engineering. Passed with honours.
- 9/2023 – 6/2024 **Master of Computational Physics (CompuPhys)** University of Besançon (France)  
Second year of Master's degree in physics, specializing in numerical physics. Passed with high honours.
- 9/2022 – 6/2023 **Master of Physics (1st year)** University of Strasbourg (France)  
Enrolled in the complementary graduate program **QMat** and participated in the 2023 edition of the **FPT (French Physicists' Tournament)**.
- 9/2021 – 5/2022 **Bachelor of Fundamental Physics** University of Strasbourg (France)  
Passed with high honours.
- 9/2019 – 5/2021 **MPA (Mathématiques et Physique Approfondies)** University of Strasbourg (France)  
Math and physics double major. Passed with honours.

## PROJECTS (see all)

- Raytracing, C++ **Ilya** [github.com/Epsylene/Ilya](https://github.com/Epsylene/Ilya)  
A basic raytracer. Currently implements different types of materials, importance sampling and a basic BVH tree.
- OpenGL, C++ **minigl** [github.com/Epsylene/minigl](https://github.com/Epsylene/minigl)  
A small OpenGL 3D graphics library.
- x86 asm, C++ **axolotl-os** [github.com/Epsylene/axolotl-os](https://github.com/Epsylene/axolotl-os)  
OS project, written in x86 assembly and C++. Currently consisting of a custom long-mode bootloader and the kernel it boots to.
- Vulkan, Rust **caliban** [github.com/Epsylene/caliban](https://github.com/Epsylene/caliban)  
An experimental Vulkan renderer written in Rust.

## LANGUAGES

**English** – C1 (TOEIC), **French** – native, **Spanish** – native